

Unveiling the Vulnerability of Graph-LLMs: An Interpretable Multi-Dimensional Adversarial Attack on TAGs

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Abstract

Graph Neural Networks (GNNs) have become a pivotal framework for modeling graph-structured data, enabling a wide range of applications from social network analysis to molecular chemistry. By integrating large language models (LLMs), text-attributed graphs (TAGs) enhance node representations with rich textual semantics, significantly boosting the expressive power of graph-based learning. However, this synergy introduces critical vulnerabilities in both topology and text. Although specialized attack methods have been designed for each of these aspects, no work has yet unified them into a comprehensive approach. In this work, we propose the Interpretable Multi-Dimensional Graph Attack (IMDGA), a human-centric framework orchestrating multi-level perturbations across graph structure and textual features. IMDGA utilizes three tightly integrated modules to craft attacks that balance interpretability and impact, enabling a deeper understanding of Graph-LLM vulnerabilities. Through rigorous theoretical analysis and comprehensive empirical evaluations on diverse datasets and architectures, IMDGA demonstrates superior interpretability, attack effectiveness, stealthiness, and robustness compared to existing methods. By exposing these underexplored semantic vulnerabilities, our work offers valuable insights for improving Graph-LLM resilience. Our code is available at <https://github.com/bwfan-bit/IMDGA>.

CCS Concepts

• Security and privacy; • Computing methodologies → Neural networks; • Information systems → Data mining;

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Keywords

Text-attributed graphs; Graph-LLMs; Adversarial attacks

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Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18309591>.

1 Introduction

In the contemporary landscape of data science, graphs are indispensable data structures for representing intricate and interactive entities [60]. Within domains such as citation networks [20, 45, 57] and social media platforms [28, 29, 58], where nodes are characteristically imbued with copious semantic content, text-attributed graphs (TAGs) distinguish themselves from traditional graphs by offering a more semantically enriched structural paradigm [7, 9, 18]. Coinciding with this, Graph Neural Networks (GNNs) [11, 34, 50] have rapidly developed into a powerful tool for modeling TAGs, effectively capturing the intricate interactions and profound semantic connections within graphs. This ability to understand both the structure and semantics of TAGs has led to outstanding performance in various downstream tasks, including biological networks [13, 42, 72] and recommendation systems [2, 19, 49].

With the ascendance of large language models (LLMs), these advanced architectures have been adapted to the graph domain, giving rise to the innovative paradigm known as Graph-LLMs [8, 22, 32, 41]. While the paradigm originated with encoder-only models (e.g., BERT [12], RoBERTa [31]), the community's focus has recently expanded to include decoder-only architectures (e.g., Llama [53]). Within this paradigm, *LLM-as-Enhancer* [7, 30] encoding scheme transcends shallow textual encoding methods (e.g., skip-gram [36]

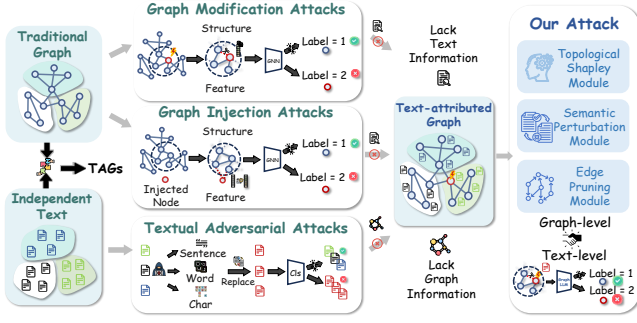


Figure 1: Illustration of different adversarial attack spaces

and BoW [17]), thereby endowing node features in TAGs with more profound semantic information. Concurrently, this non-decoupled paradigm dismantles the conventional separation of text feature processing and model architecture designing, significantly enhancing the capability of representation learning on TAGs [5, 71].

However, recent investigations have underscored the inherent vulnerability of GNNs to adversarial examples, which are meticulously crafted by introducing subtle perturbations to the original input data [15, 25, 51]. In the conventional GNN settings, attacks typically involve alterations to the graph structure or node features. Notably, Graph Modification Attacks (GMAs) [10, 52, 61, 74] and Graph Injection Attacks (GIAs) [6, 56, 68, 73], as emblematic instances, pose formidable challenges to the robustness of GNNs. In the context of TAGs, the incorporation of additional textual information into Graph-LLMs introduces new security concerns, suggesting that attack methodologies from the field of Natural Language Processing (NLP) may have a significant impact on the representation learning of existing TAGs [23, 27, 55].

While prior adversarial attack methods have explored diverse perspectives (as illustrated in Figure 1), they clearly manifest the following shortcomings in scenarios integrating LLMs with graphs:

- (1) *Incompatibility with real-world constraints*: A large portion of conventional adversarial attacks rely on either unrestricted access to the target model’s internal gradients or the deployment of poisoning data during the model training phase, both of which are often strictly unattainable in practical environments [3, 4, 47].
- (2) *Limited scope in attack paradigms*: Although GMAs and GIAs are meticulously designed attack frameworks that compromise GNNs by leveraging graph-structure knowledge, they fail to address vulnerabilities originating at the raw textual level [26]. Conversely, while textual attacks from the NLP domain can significantly mislead models in classification, the textual features on TAGs are inherently interconnected through the graph’s message-passing mechanism [59, 62], making a naive direct transfer of these methods unviable.
- (3) *Insufficient interpretability in graph attacks*: Prevailing strategies primarily focus on perturbing node features and modifying the graph topology. However, these modifications often lack a human-centric perspective and suffer from limited interpretability, hindering a deeper understanding of Graph-LLMs’ vulnerabilities and constraining the pursuit of more resilient defense approaches [21, 38].

To address these limitations, we propose the Inter-pretatable Multi-Dimensional Graph Attack (IMDGA), a black-box framework [37,

40] designed to balance effectiveness, stealth, and interpretability using only model outputs. Specifically, IMDGA operates through three progressively connected stages to achieve effective attacks on TAGs. In the **warm-up stage**, the *Topological SHAP Module* quantifies token contributions during graph propagation [33, 44, 48]. This approach illuminates the semantic weight of tokens from a human-centric perspective, pinpointing pivotal words that drive node classification. Subsequently, the **manipulation stage** employs the *Semantic Perturbation Module*. It generates context-aware substitutes via a mask model [14] and filters them using a graph-aware scoring function. This process applies optimal perturbations that disrupt predictions while preserving the attack’s stealthiness. To transcend the inherent limitations of text-only attacks, the **refinement stage** introduces the Influence-based *Edge Pruning Module* to identify the most vulnerable edges [1, 39]. It serves a dual function: it alleviates the computational bottleneck of Shapley strategies [33] and refines attack precision. By selectively pruning edges that positively reinforce classification confidence, this module significantly curtails overhead while maximizing impact. This integrated approach simultaneously enhances attack effectiveness and stealthiness, while offering valuable insights into Graph-LLM vulnerabilities.

The main contributions of our work are summarized as follows:

- ① **A Novel Perspective on TAG Security**: We present a pioneering and crucial perspective on the security vulnerabilities of TAGs, which arise from the non-decoupled nature of text encoding and GNN message passing. By synthesizing knowledge from both advanced NLP adversarial attacks and graph attack fields, we introduce a new paradigm for security analysis and attacks on TAGs.
- ② **A Unified Multi-Dimensional Attack Framework**: We propose IMDGA, a holistic framework that effectively integrates three novel modules to perform interpretable attacks on both text and edge attributes. This methodology includes a human-centric approach for targeted text substitution and, critically, an edge-level attack strategy that transcends the inherent limitations of text-only perturbations, thereby further optimizing the attack’s potency.
- ③ **Proven Effectiveness and Stealthiness**: The IMDGA method demonstrates exceptional attack success rates under stringent black-box conditions, showcasing robust performance across multiple datasets. The stealthiness of our attack is rigorously validated through empirical results and theoretical analysis, ensuring that the perturbations remain imperceptible while being highly effective.

2 Background and Preliminary

In this section, we will briefly introduce the key concepts and definitions to better explain the fundamentals of our proposed method. **Text-Attributed Graphs**. Typically, a text-attributed graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}, \mathcal{T})$, where $\mathcal{V}$ denotes the set of nodes with $|\mathcal{V}| = n$ and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents the set of edges, which can also be represented by the adjacency matrix $\mathcal{A} \in \mathbb{R}^{n \times n}$. Any node $v_i \in \mathcal{V}$ has an associated text description $\mathcal{T}_i \in \mathcal{T}$. The feature matrix $\mathcal{X} \in \mathbb{R}^{n \times d}$, derived by encoding the texts $\mathcal{T}$, contains the feature vectors for all nodes, and d represents the dimension of node features. Specifically, we define the label set as $\mathcal{Y} = \{y_1, y_2, \dots, y_n\}$, where each label y_i is uniquely associated with a node $v_i \in \mathcal{V}$.

LLM-based Graph Learning. In this section, we elucidate the workflow of graph learning on TAGs when LLMs serve as enhancers.

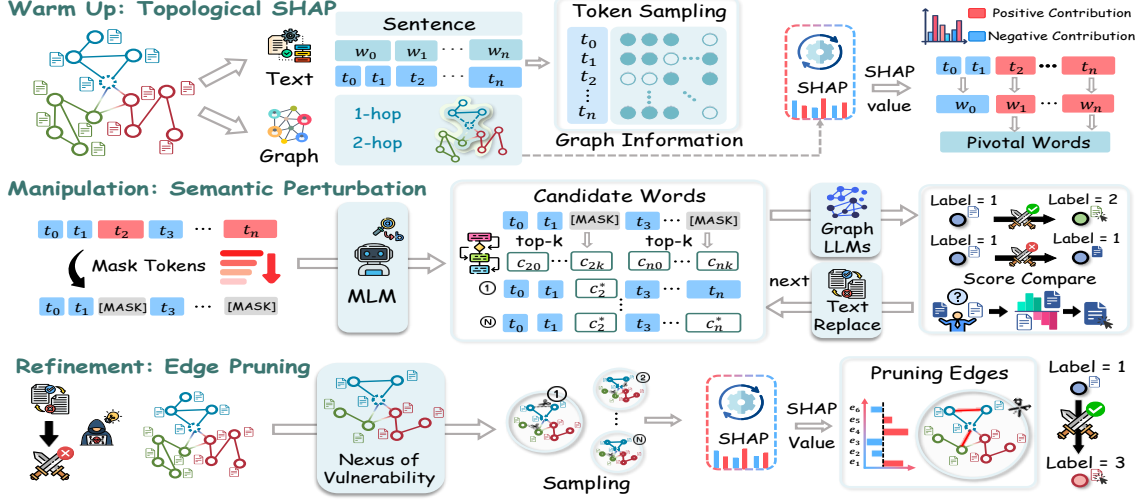


Figure 2: An overview of IMDGA framework, illustrating the key components and methodologies.

Broadly, such Graph-LLMs comprise three pivotal components: initialization, aggregation, and updating, which seamlessly integrate textual information with graph-structured data to enable efficient representation learning. For a given node v_i and its associated textual attribute t_i , the transformer-based model generally encodes the raw text into a semantically rich embedding vector to initialize the node representation, expressed as follows:

$$h_{v_i} = x_i, \quad x_i = \psi_\theta(t_i) \in \mathbb{R}^d, \quad \forall v_i \in \mathcal{V}, \quad (1)$$

where ψ_θ denotes the text encoder with parameters θ . To enrich the final representation of node v_i , update operation $f_{Up}^{(k-1)}$ and aggregation function AGGR are employed to consolidate node representations derived from the preceding iteration:

$$h_{v_i}^{(k)} = f_{Up}^{(k-1)}(h_{v_i}^{(k-1)}, \text{AGGR}(\{h_u^{(k-1)}, u \in \mathcal{N}(v_i)\})). \quad (2)$$

Embedded from raw text and then iteratively refined, these node representations capture both rich textual semantics and graph structure information, making them suitable for various downstream tasks such as node classification and link prediction.

Adversarial Attack. The vulnerability of TAGs stems from two principal attack vectors: graph and text adversarial attacks.

Graph adversarial attacks are designed to deceive a GNN $\mathcal{F}_\theta$ by subtly altering the original graph $\mathcal{G}$, causing the model to produce incorrect predictions. This objective can be formally expressed as:

$$\max \mathcal{L}(\mathcal{F}_\theta(\mathcal{G}')), \quad \text{s.t. } \|\mathcal{G}' - \mathcal{G}\| \leq \Delta, \quad (3)$$

where $\mathcal{G}'$ denotes the perturbed graph and Δ denotes the perturbation budgets, containing Δ_A and Δ_X . In GMAs, modifications are restricted to making subtle perturbations on the existing adjacency matrix $\mathcal{A}$ and feature matrix $\mathcal{X}$. Specifically, these perturbations must satisfy the constraint $\|\mathcal{A}' - \mathcal{A}\|_0 \leq \Delta_A$ and $\|\mathcal{X}' - \mathcal{X}\|_\infty \leq \Delta_X$. For GLAs, the graph structure is expanded by introducing malicious nodes $\mathcal{V}_{\text{atk}}$ with corresponding adversarial features $\mathcal{X}_{\text{atk}}$, resulting in $\mathcal{X}' = \begin{pmatrix} \mathcal{X} \\ \mathcal{X}_{\text{atk}} \end{pmatrix}$ and $\mathcal{A}' = \begin{pmatrix} \mathcal{A} & \mathcal{A}_{\text{atk}} \\ \mathcal{A}_{\text{atk}}^T & \mathcal{O}_{\text{atk}} \end{pmatrix}$. Similarly, the number of attacked nodes $|\mathcal{V}_{\text{atk}}|$, the degree of nodes d_v , and the feature matrix $\mathcal{X}_{\text{atk}}$ are bounded to ensure stealthiness.

Text adversarial attacks are commonly employed in tasks such as text classification, aiming to craft adversarial samples for the textual attributes $\mathcal{T}^*$ that significantly diminish the confidence of classifier Φ in accurate predictions, thereby inducing misclassification:

$$\mathcal{T}^* = \arg \min_{\mathcal{T}'} \{-D(\mathcal{T}, \mathcal{T}')\}, \quad \text{s.t. } \Phi(\mathcal{T}') \neq y, \quad (4)$$

where D denotes the semantic similarity. Drawing inspiration from the above adversarial attacks, we propose a novel perspective to combine their advantages, investigating the vulnerabilities of TAGs.

Shapley Value. SHAP (SHapley Additive exPlanation) is a game-theoretic framework that employs the concept of Shapley values [46, 48] to quantify the contribution of individual players (e.g., features) in cooperative scenarios. Intuitively, the more pivotal a player is to the prediction, the higher its corresponding Shapley value will be. In general, the Shapley value of a player is computed as the weighted average of all possible marginal contributions that the player provides across different coalitions S :

$$\phi(i) = \sum_{S \subseteq \{1, \dots, n\} \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [f(S \cup \{i\}) - f(S)], \quad (5)$$

where n denotes the total number of players and $f(S \cup \{i\}) - f(S)$ quantifies the marginal contribution of player i to the coalition S . From an explainable perspective, the raw text, features, or edges within TAGs can be seen as collaborative 'players' that jointly explain model predictions. This leads to the concept of Topological SHAP, which makes the attack process more interpretable.

Threat Model. We focus on untargeted evasion attacks in a strict black-box setting. This setting is restricted to probability queries, without access to model gradients or parameters. The adversary perturbs the graph structure and node texts to induce misclassification (i.e., $\mathcal{F}_\theta(\mathcal{G}')_v \neq y_v$) by minimizing the ground-truth confidence:

$$\min_{\mathcal{G}'} \mathcal{F}_\theta(\mathcal{G}')_{y_v} \quad \text{s.t. } \|\mathcal{A}' - \mathcal{A}\|_0 \leq \Delta_A, \quad \mathcal{R}(\mathcal{T}', \mathcal{T}) \leq \Delta_T, \quad (6)$$

where $\mathcal{F}_\theta(\mathcal{G}')_{y_v}$ denotes the predicted probability of the true label y_v . The structural perturbation $\|\mathcal{A}' - \mathcal{A}\|_0$ is constrained by edge budget Δ_A , and the text modification rate $\mathcal{R}(\cdot)$ is bounded by Δ_T .

3 Methodology

In this section, we present a comprehensive overview of our innovative method, which systematically probes the vulnerabilities of Graph-LLMs with a focus on interpretability. IMDGA introduces a pioneering paradigm for adversarial attacks on representation learning in TAGs under black-box conditions, achieving precise and efficient targeted manipulations. It strategically accounts for both textual attributes and graph structural dynamics in its attack design, addressing three important challenges through synergistic modules. Specifically, IMDGA first introduces the Topological SHAP to tackle **Challenge ①**: How to pinpoint pivotal words in the raw text that critically influence model predictions from a graph-centric perspective (Section 3.1)? Subsequently, it employs the Semantic Perturbation to address **Challenge ②**: How to execute stealthy, semantically coherent perturbations to these pivotal words (Section 3.2)? Finally, IMDGA proposes the Edge Pruning to achieve more advanced adversarial effects, resolving **Challenge ③**: How to precisely disrupt key message-passing pathways in a human-intuitive manner (Section 3.3)? For clarity, we illustrate the overall framework in Figure 2 and provide pseudocode in Appendix H.

3.1 Topological SHAP (Warm Up)

Words serve as the fundamental building blocks of sentences, embodying independent semantic units that often encapsulate core intent, such as subjects or sentiments. While numerous NLP techniques have pursued word-level textual attacks, the texts in TAGs differ markedly from those in prior studies as they are intrinsically tied to nodes, interconnected via edges, rendering them interdependent rather than isolated passages. Consequently, attacking a single node's text in isolation without accounting for its connected neighbors fails to fully exploit the graph's textual attributes. To address this, we introduce the Topological SHAP Module in the warm-up stage, which maps the SHAP framework onto graph structures to quantify word importance from a topological viewpoint:

$$\phi(i) = \sum_{S \subseteq \mathcal{W} \setminus \{W_i\}} \frac{|S|!(|\mathcal{W}| - |S| - 1)!}{|\mathcal{W}|!} [f(\mathcal{T}_S) - f(\mathcal{T}_{S \cup \{W_i\}})], \quad (7)$$

where for a node v , $\mathcal{W} = \{W_i \mid W_i \in \text{Tokenize}(\mathcal{T}_v), i = 1, \dots, m\}$ denotes the subset of words derived from the node's text $\mathcal{T}_v$, with W_i representing the i -th word. The masked text $\mathcal{T}_S$ is defined as:

$$\mathcal{T}_S = \{W_i \cdot \mathbb{I}_{\{W_i \notin S\}} + [\text{MASK}] \cdot \mathbb{I}_{\{W_i \in S\}} \mid W_i \in \mathcal{W}\}. \quad (8)$$

Notably, while $\mathcal{T}_S$ is defined as a set, it retains the original text sequence order when used as input. In addition, the coalitional function $f(\mathcal{T}_S)$ is defined as the aggregation of the GNN's predictive scores over v and its neighborhood $N(v)$, which serves as the basis for computing the marginal contribution:

$$f(\mathcal{T}_S) = \sum_{u \in \{v\} \cup N(v)} \mathcal{F}_\theta(\mathcal{G}, \psi_\theta(\mathcal{T}_S), u), \quad \forall v \in \mathcal{V}. \quad (9)$$

In contrast to traditional SHAP, our approach employs a mirrored operation, bypassing the conventional paradigm of incrementally adding features. By masking words, we reformulate the marginal contribution from the additive transition (S to $S \cup \{i\}$) to a subtractive shift ($S \cup \{W_i\}$ to S). The Shapley value $\phi(i)$ quantifies a word's impact on the classification of both the node and its neighborhood,

providing a precise measure of its significance in Graph-LLM predictions. Recognizing that $\phi(i)$ encompasses contributions across all classes, we refine the final score to focus on the sum of its Shapley values for the true labels of relevant nodes $\phi^{y_u}(i)$:

$$\xi(i) = \sum_{u \in \{v\} \cup N(v)} \phi^{y_u}(i), \quad \forall v \in \mathcal{V}. \quad (10)$$

Once the importance score $\xi(i)$ of each word within a sentence has been obtained, we further refine the selection to identify the most influential tokens. Accordingly, we define the pivotal word set $\mathcal{P}$ as the top- k words whose $\xi(i)$ values exceed a threshold τ :

$$\mathcal{P} = \{W_i \mid \xi(i) > \tau, i \in \mathcal{I}_k\}. \quad (11)$$

The pivotal word set $\mathcal{P}$ thus encapsulates the most semantically and topologically critical words that govern the prediction behavior of both the target node and its neighbors. By distilling the text into this compact subset of decisive words, we establish a principled foundation for manipulation stage, where carefully designed perturbations can be applied in a targeted and stealthy manner.

3.2 Semantic Perturbation (Manipulation)

Inspired by the limitations of conventional textual attacks, which fail to leverage inter-node message passing to amplify their adversarial impact through graph structures, we introduce the Semantic Substitution Module. Specifically, to generate candidate substitutes for the pivotal words in $\mathcal{P}$, this module employs a model-agnostic Masked Language Model (MLM) as a proxy to propose context-aware replacements. Operating independently of the victim model, the proxy utilizes generic pre-trained weights to capture universal linguistic patterns. Consequently, the generation process relies solely on the intrinsic probability distribution of the proxy MLM. Formally, we leverage the MLM objective to maximize the conditional likelihood of tokens given their context:

$$\prod_{i=1}^m P(W_i \mid W_1, \dots, W_{i-1}, W_{i+1}, \dots, W_m). \quad (12)$$

Building upon this principle, our framework exploits such contextual dependency modeling to synthesize semantically coherent and contextually plausible candidate substitutions. Taking node v as an example, we define its candidate word set C as the union of top- k replacements for each pivotal word $W_i \in \mathcal{P}$, where each pivotal word generates multiple semantically proximate candidates to ensure diversity and contextual adaptability. For each candidate $r \in C$, we generate the perturbed text $\mathcal{T}' = (\mathcal{T} \setminus \{W_i\}) \cup \{r\}$. Subsequently, we compute GNN's predictive probability distribution $p_u(\mathcal{G}, \mathcal{T}')$ for nodes $u \in \{v\} \cup N(v)$. The confidence gap is defined as:

$$\delta_u(r) = p_u^{(1)}(\mathcal{G}, \mathcal{T}') - p_u^{(2)}(\mathcal{G}, \mathcal{T}'), \quad (13)$$

where $p_u^{(1)}$ and $p_u^{(2)}$ denote the largest and the second-largest predicted probabilities, respectively, quantifying the model's decision certainty for node u post-replacement. Evidently, a smaller confidence gap indicates a more ambiguous decision boundary for the model, thereby rendering it more vulnerable to adversarial attacks. Therefore, aggregating these gaps across the neighborhood yields the base score for candidate word r :

$$\Delta(r) = \sum_{u \in \{v\} \cup N(v)} \delta_u(r). \quad (14)$$

To incorporate adaptive characteristics, we introduce the label-flip indicator function $\mathbb{I}_{\text{flip}}(r)$, which is 1 if the replacement induces a prediction label flip for node v or its neighborhood, and 0 otherwise. The final replacement score is defined as:

$$\sigma(r) = \Delta(r) \cdot (1 + \alpha \cdot \mathbb{I}_{\text{flip}}(r)), \quad (15)$$

where $\alpha > 0$ is a hyperparameter that dynamically amplifies the confidence gap weight in label-flip scenarios, prioritizing high-impact replacements. The aggregated score $\Delta(r)$ ensures a comprehensive neighborhood evaluation, while the adaptive factor $(1 + \alpha \cdot \mathbb{I}_{\text{flip}}(r))$ elegantly highlights boundary perturbations, balancing stability and sensitivity analysis. This innovative mechanism implicitly reinforces the graph interactions through neighborhood aggregation, endowing the attack with greater strategic depth and robustness.

3.3 Edge Pruning (Refinement)

Although textual attacks effectively disrupt node representations by perturbing high-contribution words in node texts, their efficacy is limited in scenarios where graph topology strongly dominates GNN predictions. To address this, we propose an interpretable Edge Pruning Module as a strategic extension of textual perturbations, activated only when textual disruptions fail to induce target label flips. The core of this module lies in our innovative concept, nexus of vulnerability, which represents a curated subset of nodes highly intertwined with the target node v and inherently susceptible to attacks. We identify this nexus through a robust multifaceted scoring mechanism that elegantly fuses three complementary dimensions: predictive disparity, feature influence, and vertex centrality. Drawing from the confidence gap $\delta_u(r)$ in Eq. (13), we similarly define predictive disparity $\delta(u)$ for node u , quantifying decision boundary ambiguity. Critically, feature influence is derived from message propagation dynamics, measured via the L1 norm of the expected Jacobian for node v 's impact on u after k layers:

$$I(u, v, k) = \left\| \mathbb{E} \left[(\partial X_u^{(k)}) / (\partial X_v^{(0)}) \right] \right\|_1, \quad (16)$$

where $X_u^{(k)}$ denotes node u 's feature after the k -th layer, and $X_v^{(0)}$ is node v 's initial feature embedding. However, internal gradient information is inaccessible in black-box settings. To address this, we approximate $I(u, v, k)$ using PageRank, as the expected Jacobian norm is theoretically proven to align with random walk probabilities [63]. By normalizing this approximation, we obtain:

$$I_u(v, k) = \frac{I(u, v, k)}{\sum_{w \in V} I(u, w, k)}, \quad (17)$$

reflecting node v 's relative contribution to node u 's representation. This allows us to quantify the feature influence from target node v to other nodes. Typically, nodes with lower degrees, owing to their reduced structural redundancy, are more prone to amplified perturbation effects. To capture this property, we incorporate $\frac{1}{\deg(u)}$ as a surrogate indicator of structural centrality and attack susceptibility, thereby forming the final term in the computation of the vulnerability score. We consolidate the above dimensions into a unified vulnerability scoring function, formally defined as:

$$\text{Score}(u) = \alpha_1 \cdot (1 - \delta(u)) + \alpha_2 \cdot I_u + \alpha_3 \cdot \left(\frac{1}{\deg(u)} \right), \quad (18)$$

Nodes with the highest scores form the nexus of vulnerability $\mathcal{G}_n(v)$, constraining the attack domain to a high-impact subgraph. On this basis, we delineate critical paths linking nodes in $\mathcal{G}_n(v)$ to v , and employ an efficient Shapley value approximation inspired by GNNShap, cast as a least-squares solution:

$$\hat{\phi} = (M^T U M)^{-1} M^T U \hat{y}, \quad (19)$$

where $M \in \mathbb{R}^{k \times n}$ is the mask matrix encoding k subgraph samples across n edges, U is a diagonal weight matrix reflecting sample importance and $\hat{y}$ approximates nexus predictions under masked configurations. Finally, we prune the top- k edges with the highest attributions, further eroding model confidence in node's original label. This interpretable pruning mechanism not only complements textual attacks but also uncovers topological vulnerabilities, forging a multi-dimensional adversarial strategy that advances the frontier of explainable TAG attacks.

3.4 Time Complexity Analysis

To substantiate the scalability of IMDGA, we provide a theoretical analysis highlighting how interpretability is achieved without sacrificing computational efficiency. First, the exponential cost of exact Shapley computation is alleviated through partition-based sampling; for a node with m tokens, the coalition space is reduced to size $s \ll 2^m$. With t_g denoting the inference time of the underlying Graph-LLM, the complexity of the Topological SHAP Module becomes $O(s \cdot t_g)$. For token substitution, the Semantic Perturbation Module bounds computation by restricting the search to $|\mathcal{P}|$ pivotal tokens, each producing k_c semantically coherent candidates; since the masked language model requires only a single forward pass, the cost is $O(k_c |\mathcal{P}| \cdot t_g + t_m)$, where t_m is the MLM inference time. Finally, the Edge Pruning Module narrows the sampling space to a subgraph $\mathcal{G}_n(v)$ around the target node and further samples $k \ll 2^{|\mathcal{N}(v)|}$ coalitions within this restricted domain, which not only limits the computational scope but also improves the precision of Shapley-based attribution, yielding a complexity of $O(k \cdot t_g)$. Integrating these components, the overall per-node complexity converges to $O((s + k + k_c |\mathcal{P}|) \cdot t_g + t_m)$, demonstrating that the complexity of IMDGA increases slowly with the scale of the dataset, being primarily constrained by the underlying Graph-LLMs and the chosen sampling hyperparameters.

4 Experiments

In this section, we conduct a comprehensive evaluation of the proposed IMDGA framework to demonstrate its effectiveness, interpretability, stealthiness, and robustness on TAGs. To rigorously assess these attributes, we structure our experiments around the following research questions: **Q1**: To what extent does IMDGA method surpass traditional approaches in executing effective attacks on TAGs? **Q2**: Can our method optimally reconcile the trade-off between attack stealthiness and effectiveness? **Q3**: Does IMDGA maintain robust attack performance when facing different Graph-LLM architectures? **Q4**: What distinct roles do the individual modules play in bolstering the overall performance of the IMDGA framework? **Q5**: How significantly do critical hyperparameters influence the robustness and adaptability of our attack strategy on TAGs?

Table 1: Comprehensive Comparison of Adversarial Attack Effectiveness Across TAG Datasets.

Dataset	Methods	SBERT		BERT		RoBERTa		DeBERTa		DistilBERT	
		ACC ↓	ASR ↑	ACC ↓	ASR ↑	ACC ↓	ASR ↑	ACC ↓	ASR ↑	ACC ↓	ASR ↑
Cora	Clean	82.14±1.71	-	80.61±1.85	-	77.48±0.91	-	76.19±1.16	-	81.31±0.06	-
	HLBB	76.97±0.98	23.33±0.53	76.42±0.86	32.41±1.94	73.70±1.05	24.54±0.36	74.80±1.06	9.72±1.07	78.22±1.15	23.15±0.78
	TextHoaxer	78.45±1.35	16.67±0.61	76.83±1.92	31.16±0.08	74.25±1.17	22.22±0.55	74.90±1.04	9.68±1.88	79.04±0.33	18.52±1.49
	SemAttack	81.13±0.29	4.44±0.15	79.42±0.91	6.05±1.14	76.65±0.58	5.09±0.13	75.36±0.36	6.48±1.07	80.30±1.96	6.48±0.43
	FGSM	71.02 ±1.63	43.70±0.49	70.37±1.97	57.21±0.11	64.61±1.42	63.43±0.78	69.17±0.25	45.37±1.51	70.47±0.89	61.11±1.06
	PA-F	63.68 ±1.22	70.74±0.76	71.94±1.91	46.05±0.37	64.28±0.85	61.11±1.45	67.60±0.04	48.61±1.19	75.50±1.58	37.96±0.52
	IMDGA	61.51±1.73	95.19±0.02	64.33±1.48	94.42±0.87	61.98±0.55	94.44±1.94	61.24±0.33	95.83±1.21	65.25±0.69	93.06±1.08
Citeseer	Clean	71.58±1.16	-	72.54±0.50	-	71.48±0.82	-	70.26±0.91	-	72.85±0.37	-
	HLBB	64.14±1.05	43.08±0.67	67.28±1.93	40.16±0.18	69.16±1.34	18.50±0.42	66.42±0.27	26.38±1.58	69.48±1.77	29.13±0.96
	TextHoaxer	66.65±1.41	27.36±0.98	69.75±0.25	23.23±1.80	69.83±0.07	14.57±1.63	68.73±0.56	11.02±1.97	70.46±0.39	20.47±1.11
	SemAttack	70.26±1.25	7.23±0.78	71.32±1.91	8.66±0.33	71.17±0.62	3.15±1.06	68.07±1.54	15.35±0.09	71.95±1.77	9.06±0.48
	FGSM	59.83 ±1.04	58.81±0.61	64.26±1.98	51.97±0.15	61.91±1.42	61.02±0.73	63.52±0.36	43.31±1.55	64.73±0.89	55.51±1.22
	PA-F	68.77±1.33	14.15±0.69	69.95±1.92	19.29±0.18	62.18±1.04	58.27±0.77	61.75±0.25	55.51±1.55	69.16±0.48	27.95±1.87
	IMDGA	53.82±1.04	92.77±0.61	58.65±1.98	94.49±0.15	57.94±1.42	93.31±0.73	56.89±0.36	93.31±1.55	59.40±0.89	94.10±1.22
PubMed	Clean	82.31±1.45	-	82.31±0.78	-	83.71±0.29	-	82.62±1.94	-	82.04±0.05	-
	HLBB	81.12±1.73	27.67±0.02	81.25±1.48	30.00±0.87	82.95±0.55	25.80±1.94	82.03±0.33	18.60±1.21	81.29±0.69	23.80±1.08
	TextHoaxer	81.15±1.19	28.67±0.58	81.27±1.92	28.20±0.03	82.88±1.35	25.00±0.81	82.03±0.46	16.80±1.64	81.33±0.99	22.00±1.87
	SemAttack	81.93±1.04	16.00±0.61	82.10±1.98	5.40±0.15	83.54±1.42	5.40±0.73	82.43±0.36	17.20±1.55	81.97±0.89	4.00±0.22
	FGSM	80.27±1.19	64.33±0.58	78.65 ±1.92	69.00±0.03	80.67±1.35	72.40±0.81	81.67±0.46	35.20±1.64	78.59±0.99	57.40±1.87
	PA-F	80.20±1.04	13.29±0.61	80.76±1.98	29.40±0.15	80.65±1.42	51.00±0.73	81.70±0.36	32.60±1.55	80.67±0.89	17.80±1.22
	IMDGA	77.62±1.19	85.88±0.58	77.84±1.92	86.40±0.03	80.17±1.35	84.40±0.81	79.35±0.46	77.40±1.64	77.88±0.99	86.00±1.87

4.1 Experimental Setup

Datasets. We evaluate our method on three standard citation datasets (Cora, Citeseer, PubMed) [64] and the large-scale ogbn-arxiv dataset [20]. Detailed statistics are provided in Appendix A.

Compared Baselines. To ensure a fair and comprehensive comparison of adversarial performance on TAGs, we select baselines from both the text adversarial attack domain and the graph adversarial attack domain (see Appendix B). From the former category, we select HLBB [37], TextHoaxer [65], and SemAttack [55], while additionally including LimeAttack [70] in Appendix E. These methods are adapted to TAGs to evaluate their transferability. From the latter, we adopt FGSM [15] and PA-F [35], which enable a direct evaluation against graph-specific perturbations.

Evaluation. We evaluate attack effectiveness from two perspectives: Global Accuracy (ACC) on the test set to measure overall model performance degradation, and Local Attack Success Rate (ASR) on originally correctly classified nodes to assess targeted disruption. Detailed settings are presented in Appendix C.

4.2 Performance Comparison (Q1)

Our experimental results consistently demonstrate the superior effectiveness and robustness of the proposed method across both local and global perspectives, substantially outperforming existing baselines and exposing the vulnerability of Graph-LLMs to adversarial manipulations. Through comprehensive data analysis, we derive the following key observations: (1) **Comprehensive SOTA performance:** At the local level, our strategy consistently surpasses all competing methods, producing the strongest disruptive effect on target nodes across every dataset and Graph-LLM variant (Table 1). On smaller benchmarks such as Cora and Citeseer, the ASR exceeds

90%, showing that nearly all targeted nodes can be reliably compromised. Even on larger-scale datasets like ogbn-arxiv, our method maintains clear superiority. Furthermore, with Llama-2-7b [53] as the encoder, IMDGA consistently outperforms the competitive baseline. Extended results are provided in Appendix E. From the global perspective, although the degree of degradation is influenced by factors such as node selection strategy and perturbation scale, under a unified random selection protocol our approach still achieves significantly stronger global disruption than any baseline. (2) **Overcoming prior limitations:** Traditional NLP adversarial attacks, despite their success in text classification, exhibit severely diminished transferability in the context of TAGs. Even in its best case, HLBB achieves only 40.16% ASR on Citeseer (with BERT encoding), while in other scenarios certain methods yield ASR values dropping below 10%. Meanwhile, graph-specific adversarial baselines such as FGSM and PA-F perform better under the same similarity constraints, but remain limited to embedding-level manipulations without semantic interpretability. By contrast, our method not only achieves stronger quantitative performance but also introduces meaningful and explainable perturbations, thereby addressing the twofold shortcomings of insufficient transferability in text-based attacks and lack of interpretability in graph-based ones.

4.3 Stealthiness Evaluation (Q2)

In this section, we evaluate adversarial stealthiness across two perspectives aligned with IMDGA’s modules. For the textual perspective, we select BERT cosine similarity (Sim), GPT-2 perplexity (PPL) [43], and human ratings (see Appendix F) as our main metrics. From the structural perspective, we report both degree distributions and the feature-cosine homophily [6]. Specifically, the

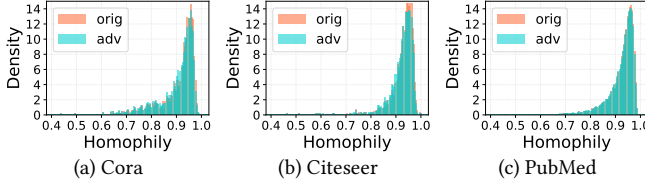


Figure 3: Feature-cosine homophily (before and after attack).

Table 2: Textual stealthiness.

Method	Cora			Citeseer		
	PPL	Sim	Human	PPL	Sim	Human
HLBB	638.35	0.917	3.254	574.46	0.934	3.391
TextHoaxer	618.55	0.914	3.290	580.38	0.955	3.326
SemAttack	498.46	0.954	3.486	209.99	0.958	3.493
IMDGA	196.74	0.955	3.565	186.18	0.960	3.529

Table 3: Generalizable Robustness and Effectiveness of IMDGA Across Multiple Backbones (ASR).

Encoder	Backbone	Cora	Citeseer	PubMed
SBERT	GCN	92.19 ± 0.84	94.87 ± 0.42	87.22 ± 1.02
	GAT	83.58 ± 0.67	93.48 ± 1.35	90.72 ± 0.64
	SAGE	95.21 ± 1.89	92.74 ± 0.58	88.79 ± 1.05
	RGCN	89.04 ± 0.35	91.19 ± 1.81	84.66 ± 0.92
	Guard	93.92 ± 1.76	98.12 ± 0.36	90.12 ± 1.58
	GLEM	88.73 ± 0.54	94.55 ± 0.43	90.11 ± 0.47
	GIANT	87.19 ± 0.98	93.67 ± 1.24	84.39 ± 1.26
BERT	GCN	92.37 ± 1.42	95.63 ± 0.72	85.42 ± 1.37
	GAT	90.12 ± 0.74	93.26 ± 1.18	91.73 ± 0.72
	SAGE	89.77 ± 1.63	92.41 ± 0.66	89.04 ± 0.92
	RGCN	87.03 ± 0.42	91.86 ± 1.64	85.13 ± 1.08
	Guard	86.47 ± 1.59	97.89 ± 0.41	91.28 ± 1.33
	GLEM	89.21 ± 0.61	94.32 ± 0.49	90.54 ± 0.53
	GIANT	87.68 ± 1.05	93.24 ± 1.36	84.92 ± 1.12

feature-cosine homophily of each node u is formally defined as:

$$h_u = \cos(r_u, x_u), \quad r_u = \sum_{j \in \mathcal{N}(u)} \frac{1}{\sqrt{d_u d_j}} x_j, \quad (20)$$

where $\mathcal{N}(u)$ is the neighbor set, d_u denotes degree, and x_u is the feature vector. Term r_u represents normalized neighbor aggregation. Small deviations in Δh_u and Δd_u imply high stealthiness.

As shown in Table 2, IMDGA outperforms text-only baselines in Sim and PPL, corroborated by human ratings (Appendix F). This superior preservation stems from the *Semantic Substitution Module*, which filters candidates via MLM contexts (Eq. (12)) and neighborhood confidence gaps (Eqs. (13)–(15)). Analytically, replacing a fraction ρ of tokens under an MLM similarity threshold γ yields:

$$\|x' - x\| \leq \rho \sqrt{2(1 - \gamma)} \Rightarrow \cos(x', x) \geq 1 - \rho^2(1 - \gamma). \quad (21)$$

This inequality bounds similarity for small ρ and large γ . Combined with adaptive scoring (Eq. (15)) which favors high-likelihood

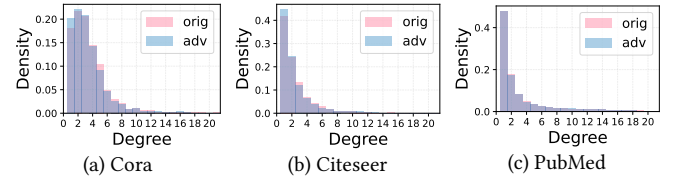


Figure 4: Degree distributions (before and after attack).

candidates to limit PPL growth, the module yields impactful yet linguistically plausible perturbations.

Post-attack homophily distributions align closely with the originals (Figure 3). This alignment arises from the strict pruning budget (top- k in Eq. (19)) and degree-normalized aggregation (Eq. (20)). Theoretically, deleting an edge (u, v) incurs a bounded shift:

$$|\cos(r'_u, x_u) - \cos(r_u, x_u)| \lesssim \frac{1}{\sqrt{d_u d_v}} \cdot \frac{\|x_v\|}{\|r_u\|}, \quad (22)$$

The degree distributions, as shown in Figure 4, align with the original curves, confirming that the global topology is preserved. Let $|\Delta E|$ denote the number of pruned edges. Since removing an edge reduces the degrees of two nodes, the mean global shift is:

$$\frac{1}{|V|} \sum_u |d'_u - d_u| = \frac{2|\Delta E|}{|V|}. \quad (23)$$

This shift is negligible because the attack is confined to the vulnerability nexus $G_n(v)$ (Eq. (18)), which limits the pruning budget.

4.4 Attack Robustness Assessment (Q3)

Having demonstrated the effectiveness and stealthiness of IMDGA in previous sections, we further investigate its generalization and robustness through extensive experiments. To verify that our attack is not limited to a specific GNN architecture, we evaluate IMDGA on widely-used models such as GCN [24], GAT [54], and GraphSAGE [16], as well as on more robust architectures including RGCN [69] and GNNGuard [66]. For Graph-LLMs, although Q1 compared multiple encoder models, the attack performance on more advanced Graph-LLMs remains unexplored. Therefore, we conduct comprehensive and rigorous experiments using GLEM [67] and GIANT [9] as representative Graph-LLMs.

The results presented in Table 3 indicate that all tested models exhibit relatively weak resistance to IMDGA, with only minor differences in ASR across architectures. Notably, even the robust Guard model achieves an ASR of 98.12% on Citeseer, suggesting that while these backbones leverage node similarity to enhance robustness, they remain vulnerable to attacks exploiting textual information. Similarly, GLEM and GIANT, which employ fine-tuned encoders to incorporate richer node representations, do not demonstrate improved resistance against our proposed adversarial attack. These findings collectively highlight that IMDGA consistently compromises diverse GNNs and Graph-LLMs, underscoring the persistent vulnerabilities present across modern graph-based models.

4.5 Ablation Study (Q4)

To evaluate the contribution of each component in our adversarial framework, we conduct ablation experiments and report the results in Table 4. For the Topological SHAP Module, we replace the

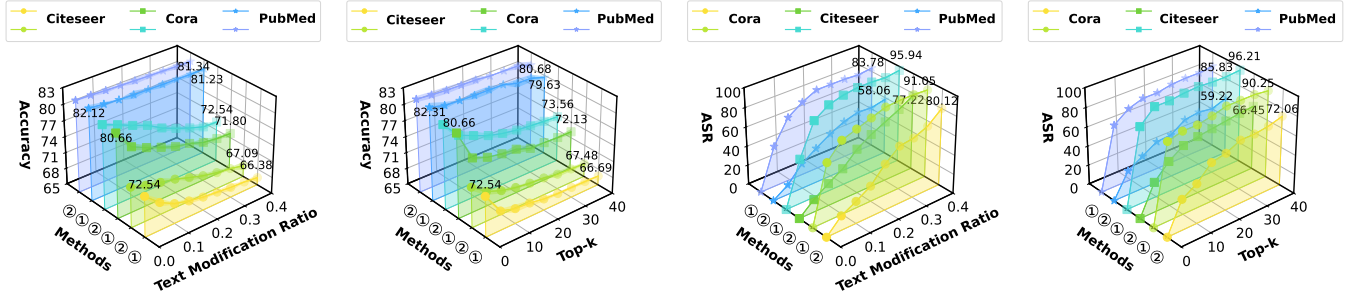


Figure 5: Sensitivity of IMDGA (① represents the full IMDGA and ② denotes text-only perturbations).

Table 4: Ablation study (ASR).

Model	Cora	Citeseer	PubMed
IMDGA (SBERT)	91.78±0.72	94.86±0.79	85.47±0.41
w/o Topology SHAP	29.54±0.42	32.89±0.29	23.76±0.58
w/o Semantic Perturb	36.97±0.53	41.21±0.19	32.84±0.44
w/o Edge Pruning	66.12±0.35	62.47±0.33	59.21±0.55
IMDGA (BERT)	92.06±0.61	95.11±0.87	85.16±0.35
w/o Topology SHAP	29.13±0.38	33.18±0.21	23.41±0.63
w/o Semantic Perturb	37.32±0.46	40.93±0.16	33.21±0.38
w/o Edge Pruning	65.87±0.30	62.23±0.26	58.83±0.62

identified pivotal words with randomly selected words for comparison. For the Semantic Perturbation Module, while it is infeasible to completely remove the effect of the MLM, we substitute the graph-aware scoring function originally designed to integrate both structural and textual information with a simpler function that only considers the prediction probability drop of a single node. For the Edge Pruning Module, we directly eliminate it, leaving only text-based perturbations in the algorithm. It is worth noting that, to preserve stealthiness, we did not apply the Edge Pruning strategy when ablating the first two modules.

From the analysis of results, several conclusions can be drawn: (1) Text modification modules are critical to attack success. When perturbations are restricted to naive random word substitutions or applied without incorporating graph structural information, the ASR sharply drops from around 92% to below 40%. Such simplified attacks are insufficient to substantially compromise the robustness of Graph-LLMs. By contrast, precisely identifying pivotal words in message passing and applying carefully designed graph-aware substitutions leads to a stronger adversarial influence on TAGs. (2) Word-level perturbations alone face inherent limitations. Even when pivotal words are correctly identified and semantically reasonable substitutions are introduced, the ASR remains capped at around 60%. Without the complementary structural guidance provided by edge-level manipulations, attacks struggle to overcome the inherent robustness of representation learning in TAGs.

4.6 Hyperparameter Analysis (Q5)

To address Q5, we conducted extensive experiments focusing on two pivotal hyperparameters: the text modification ratio β and the top- k selection parameter. These two factors largely govern the strength of adversarial perturbations and thus serve as the most

representative indicators for sensitivity analysis. In contrast, other hyperparameters play a more secondary role and are discussed in detail, along with their corresponding search spaces, in Appendix D. For experimental efficiency and comparability, we randomly selected 100 nodes as target instances in each trial, ensuring stable and reliable evaluation across different settings.

The results, summarized in Figure 5, reveal a consistent and interpretable trend: as β and top- k increase, both ACC and ASR undergo a rapid initial change, followed by a clear attenuation in marginal gains. This observation suggests that moderate relaxation of these constraints brings substantial improvements in attack effectiveness at first, as models quickly become more vulnerable under stronger perturbations. However, further increases lead to diminishing returns, with only marginal benefits despite greater perturbation. More critically, loosening these hyperparameters significantly raises the risk of exposure, as excessive modifications are more detectable, while also incurring higher computational and temporal overhead. Consequently, there exists an inherent trade-off among effectiveness, stealth, and efficiency when choosing hyperparameters. Practically, this trade-off implies that optimal settings for β and top- k should be chosen according to the deployment scenario—prioritizing higher stealth for stealth-sensitive applications and higher strength where maximal disruption is required.

5 Conclusion

In this work, we introduced IMDGA to investigate the heightened vulnerabilities of TAGs induced by the integration of textual features into Graph-LLMs. By jointly leveraging Topological SHAP, Semantic Perturbation, and Edge Pruning, IMDGA orchestrates multi-layered adversarial manipulations that expose weaknesses in both textual and structural dimensions. Through extensive evaluations, our method consistently outperforms conventional NLP and graph adversarial attack baselines in terms of interpretability, effectiveness, and stealthiness, achieving high ASR across diverse datasets. These findings not only uncover fundamental fragilities in TAG representation learning but also underscore the urgent need for systematic defenses to guide the development of more resilient Graph-LLMs against increasingly sophisticated adversarial threats.

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A Datasets

Experiments are carried out on several widely used, representative text-attributed graph datasets (Table 5). In these datasets, each node is associated with a short textual description (e.g., paper title or abstract, product description), and edges encode citations. Node features are obtained by encoding node texts with our encoding model. We adopt stratified splits of 10%/10%/80% (train/val/test) for Cora, Citeseer, and PubMed, and 20%/20%/60% for ogbn-arxiv.

Table 5: Dataset statistics.

Dataset	#Nodes	#Edges	#Classes	Avg. Degree
Cora	2,708	10,556	7	3.90
Citeseer	3,186	8,450	6	2.65
PubMed	19,717	88,648	3	4.50
ogbn-arxiv	169,343	2,315,598	40	13.67

B Attack Methods

Two families of baselines are considered on text-attributed graphs: text-side attacks that edit node texts and feature-side attacks that perturb continuous text-derived features. For fair and stealthy comparison, all methods attack the same pre-specified nodes under an untargeted setting with matched cosine-similarity thresholds.

HLBB [37] is a hard-label, decision-based black-box attack that observes only the predicted label and uses a population-based word-substitution search to push the example across the decision boundary while preserving semantic similarity.

TextHoaxer [65] is a budget-aware hard-label attack that casts discrete word substitution as a continuous optimization in embedding space. It iteratively refines a single candidate with a loss combining semantic similarity, pairwise perturbation, and sparsity to reduce queries while maintaining fluency.

SemAttack [55] is a semantics-preserving attack that defines discrete perturbations across multiple diverse semantic spaces (typos, WordNet synonyms, and contextualized BERT neighborhoods) and effectively optimizes candidates within these spaces while strictly controlling the magnitude of semantic changes.

LimeAttack [70] is a decision-based hard-label attack that leverages local interpretability to approximate word importance rankings, bypassing the inherent limitations of random initialization. It subsequently employs a beam search strategy to optimize discrete adversarial perturbations, effectively generating successful examples under a restricted query budget.

PA-F [35] is a black-box baseline that perturbs continuous node features while keeping the graph structure fixed. The original method includes RWCS-based node selection with a GC-RWCS variant. In our setting, we disable node selection and attack a fixed node set, applying only feature perturbation. Perturbations are further bounded by a cosine-similarity threshold for fairness.

FGSM [15] is a one-step gradient-sign attack in the continuous feature space. Given features x , it computes $\eta = \epsilon \text{sign}(\nabla_x J)$ and projects $x' = \Pi_C(x + \eta)$ onto a feasible set that enforces either an ℓ_∞ or an ℓ_2 bound together with a cosine-similarity threshold. The graph topology and raw tokens remain fixed.

C Experimental Settings

In this section, we present a unified description of the experimental setups for each problem to ensure reproducibility and consistency. To guarantee the reliability of the results, each experiment is repeated five times independently, and both the mean and standard deviation are reported. For Q1, we select 10% of nodes from Cora and Citeseer as target nodes, while for the larger datasets PubMed and ogbn-arxiv, we sample 500 nodes. Regarding the key parameters, the text perturbation ratio and the candidate set size are fixed at 30% and top-30, respectively. It is worth noting that since PA-F and FGSM perturb the original features, we constrain them using the same similarity measure as IMDGA. For the remaining experiments, to improve efficiency, the number of target nodes is fixed at 100, while all other hyperparameters remain unchanged. Unless otherwise specified, we choose a 2-layer GCN as the backbone.

D Hyperparameter Settings

Table 6 details the hyperparameters and search ranges used in all reported experiments, with notation following the main text.

Table 6: Search Space for IMDGA.

Hyperparameter	Description	Search Space
β	Text Modification Ratio	$\{0, 0.05, \dots, 0.4\}$
α	Label Flip Weight	$\{0, 1, \dots, 5\}$
top- k_1	Candidate Word Number	$\{0, 5, \dots, 40\}$
top- k_2	Edge Pruning Number	$\{0, 2, 4, 6\}$
$\alpha_1, \alpha_2, \alpha_3$	Scoring Function Weight	Grid Search

E Extended Experiments

Table 7 reports additional ASR results on ogbn-arxiv under identical budgets and cosine-similarity constraints. **OOT**: out of time (12-h wall-clock limit exceeded).

Table 7: ASR comparison on ogbn-arxiv.

Dataset	Methods	SBERT	BERT	RoBERTa	DeBERTa	DistilBERT
Arxiv	HLBB	35.12 $\pm$ 0.32	21.76 $\pm$ 0.84	18.14 $\pm$ 1.57	39.76 $\pm$ 0.49	37.16 $\pm$ 1.22
	TextHoaxer	36.41 $\pm$ 1.15	22.24 $\pm$ 0.82	22.39 $\pm$ 0.43	40.03 $\pm$ 0.11	38.83 $\pm$ 1.56
	SemAttack	OOT	OOT	OOT	OOT	OOT
	FGSM	30.44 $\pm$ 0.26	35.09 $\pm$ 0.88	18.81 $\pm$ 1.07	31.87 $\pm$ 0.70	30.55 $\pm$ 1.92
	PA-F	4.37 $\pm$ 0.42	21.13 $\pm$ 1.16	15.27 $\pm$ 0.51	4.56 $\pm$ 0.67	5.62 $\pm$ 0.45
	IMDGA	47.82 $\pm$ 1.03	45.56 $\pm$ 0.22	47.32 $\pm$ 0.48	44.73 $\pm$ 0.11	48.12 $\pm$ 0.45

Furthermore, we extend our evaluation to decoder-only LLMs by adopting Llama-2-7b [53] as the text encoder. In this setting, we compare IMDGA with the advanced LimeAttack baseline. As shown

in Table 8, our method maintains a clear advantage, consistently achieving higher ASR and yielding lower ACC across all datasets.

Table 8: Attack Effectiveness with Llama-2-7b.

Encoder	Methods	Cora		Citeseer	
		ACC $\downarrow$	ASR $\uparrow$	ACC $\downarrow$	ASR $\uparrow$
Llama-2	Clean	76.23 $\pm$ 1.43	–	73.48 $\pm$ 1.10	–
	LimeAttack	74.17 $\pm$ 0.58	13.02 $\pm$ 0.41	71.10 $\pm$ 1.43	15.02 $\pm$ 0.92
	IMDGA	65.45 $\pm$ 0.11	68.14 $\pm$ 0.33	60.99 $\pm$ 1.64	78.98 $\pm$ 0.49

F Human Evaluation

We recruit three trained student annotators to evaluate text-side adversarial examples after a short warm-up. For each case, sentences are shown in random, blind order, and each annotator rates them independently without discussion. Scores are averaged per sentence across annotators and then combined into per-method means across all cases for reporting.

Below are the annotation instructions and the concise guideline for language-quality ratings used herein.

Please rate overall language quality on a 1–5 scale (coherence, fluency, grammar), considering clarity and readability.

5 – Natural, coherent; no errors; fully fluent.

4 – Minor issues; easy to read; few typos.

3 – Clear meaning; some roughness; occasional errors.

2 – Awkward; hinders understanding; frequent errors.

1 – Incoherent; severely ungrammatical; unreadable.

G Adversarial Examples

Table 9: Examples of successful IMDGA attacks on Cora: original tokens in bold and adversarial substitutions in red.

Type	Text
Orig.	Prior information and generalized questions: This paper ... uses a Bayesian decision theoretic framework , contrasting parallel and inverse decision problems, ... a subsequent risk minimization ...
Adv.	Prior knowledge and simplified questions: This paper ... uses a Bayesian decision theoretic system , contrasting parallel and opposite decision problems, ... a subsequent cost minimization ...
Orig.	Several computer algorithms for discovering patterns in groups of protein sequences ... and these algorithms are sometimes prone to producing models that are incorrect because two or ...
Adv.	Several computer algorithms for discovering patterns in sets of protein sequences ... and these methods are sometimes vulnerable to producing models that are inaccurate because two or ...

H Pseudo-code

For completeness and clarity, the full procedures of the three IMDGA modules are provided in Algorithms 1–3 below.

Algorithm 1: Topological SHAP Module

Input : Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}, \mathcal{T})$, target node $v \in \mathcal{V}$, victim Graph-LLM $\mathcal{F}_\theta(\cdot)$, text encoder $\psi_\theta(\cdot)$, mask operation $Mask(\cdot)$, coalition sampler $\mathcal{S}(\cdot)$, node tokens $\mathcal{W}$, number of tokens m

Output : Pivotal word set $\mathcal{P}$

```

1 foreach  $W_i \in \mathcal{W}$  do
2   foreach  $S \in \mathcal{S}(\mathcal{W} \setminus \{W_i\})$  do
3      $\mathcal{T}_{S \cup \{W_i\}} \leftarrow Mask(S \cup \{W_i\}), \quad \mathcal{T}_S \leftarrow Mask(S)$ 
4      $z_S \leftarrow \psi_\theta(\mathcal{T}_S), \quad z_{S \cup \{W_i\}} \leftarrow \psi_\theta(\mathcal{T}_{S \cup \{W_i\}})$ 
5      $f(\mathcal{T}_S) \leftarrow \sum_{u \in \{v\} \cup \mathcal{N}(v)} \mathcal{F}_\theta(\mathcal{G}, z_S, u)$ 
6      $f(\mathcal{T}_{S \cup \{W_i\}}) \leftarrow \sum_{u \in \{v\} \cup \mathcal{N}(v)} \mathcal{F}_\theta(\mathcal{G}, z_{S \cup \{W_i\}}, u)$ 
7      $w_S \leftarrow \frac{|S|! (m - |S| - 1)!}{m!}$ 
8      $\phi(W_i) \leftarrow \phi(W_i) + w_S \cdot (f(\mathcal{T}_S) - f(\mathcal{T}_{S \cup \{W_i\}}))$ 
9 foreach  $W_i \in \mathcal{W}$  do
10    $\xi(W_i) \leftarrow \sum_{u \in \{v\} \cup \mathcal{N}(v)} \phi^{y_u}(W_i)$ 
11  $\mathcal{I}_k \leftarrow$  indices of top- $k$  tokens by  $\xi(\cdot)$ 
    /* Select top- $k$  pivotal tokens */
12  $\mathcal{P} \leftarrow \{W_i \mid \xi(W_i) > \tau, i \in \mathcal{I}_k\}$ 
13 return  $\mathcal{P}$ 

```

Algorithm 2: Semantic Perturbation Module

Input : Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}, \mathcal{T})$, target node $v \in \mathcal{V}$, Graph-LLM $\mathcal{F}_\theta(\cdot)$, text encoder $\psi_\theta(\cdot)$, Masked Language Model (MLM), pivotal word set $\mathcal{P}$, number of tokens m

Output : Perturbed text $\mathcal{T}'_v$ for node v

```

1  $C \leftarrow \{\}, \quad count \leftarrow 0, \quad \mathcal{T}'_v \leftarrow \mathcal{T}_v$ 
2 for  $i \leftarrow 0$  to  $|\mathcal{P}| - 1$  do
3    $W_i \leftarrow \mathcal{P}[i]$ 
4    $C[i] \leftarrow$  top- $k$  candidates from MLM( $W_i, \mathcal{T}_v$ )
5 for  $i \leftarrow 0$  to  $|\mathcal{P}| - 1$  do
6    $W_i \leftarrow \mathcal{P}[i]$ 
7   foreach  $r \in C[i]$  do
8      $\tilde{\mathcal{T}} \leftarrow (\mathcal{T}'_v \setminus \{W_i\}) \cup \{r\}$ 
9      $p_u(\mathcal{G}, \tilde{\mathcal{T}}) \leftarrow \mathcal{F}_\theta(\mathcal{G}, \psi_\theta(\tilde{\mathcal{T}}), u)$ 
10     $\delta_u(r) \leftarrow p_u^{(1)}(\mathcal{G}, \tilde{\mathcal{T}}) - p_u^{(2)}(\mathcal{G}, \tilde{\mathcal{T}})$ 
11     $\Delta(r) \leftarrow \sum_{u \in \{v\} \cup \mathcal{N}(v)} \delta_u(r)$ 
12    if  $\mathcal{F}_\theta(\mathcal{G}, \psi_\theta(\tilde{\mathcal{T}}), v)$  flips label then
13       $\mathbb{I}_{flip}(r) \leftarrow 1$ 
14    else
15       $\mathbb{I}_{flip}(r) \leftarrow 0$ 
16     $\sigma(r) \leftarrow \Delta(r) \cdot (1 + \alpha \cdot \mathbb{I}_{flip}(r))$ 
17     $r^* \leftarrow \arg \max_{r \in C[i]} \sigma(r)$ 
18     $\mathcal{T}'_v \leftarrow (\mathcal{T}'_v \setminus \{W_i\}) \cup \{r^*\}$ 
19     $count \leftarrow count + 1$  if  $count > \beta \cdot m$  then return  $\mathcal{T}'_v$ 
    /* Stop if modification exceeds ratio */
20 return  $\mathcal{T}'_v$ 

```

Algorithm 3: Edge Pruning Module

Input : Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}, \mathcal{T}')$, target node $v \in \mathcal{V}$, victim Graph-LLM $\mathcal{F}_\theta(\cdot)$, text encoder $\psi_\theta(\cdot)$, number of layers k , mask matrix $M \in \mathbb{R}^{k \times n}$, weight matrix U , masked prediction vector $\hat{y}$

Output : Pruned graph $\mathcal{G}'$

Parameter : weights $\alpha_1, \alpha_2, \alpha_3$

```

1  $\mathcal{G}_n(v) \leftarrow \emptyset$ 
2 foreach  $u \in \mathcal{V}$  do
3    $\delta(u) \leftarrow p_u^{(1)}(\mathcal{G}, \mathcal{T}_v') - p_u^{(2)}(\mathcal{G}, \mathcal{T}_v')$ 
4    $I(u, v, k) \leftarrow \left\| \mathbb{E} \left[ (\partial \mathcal{X}_u^{(k)}) / (\partial \mathcal{X}_v^{(0)}) \right] \right\|_1$ 
5    $I_u(v, k) \leftarrow \frac{I(u, v, k)}{\sum_{w \in \mathcal{V}} I(u, w, k)}$ 
6    $Score(u) \leftarrow \alpha_1 \cdot (1 - \delta(u)) + \alpha_2 \cdot I_u(v, k) + \alpha_3 \cdot \left( \frac{1}{\deg(u)} \right)$ 
    /* Calculate the weighted score of predictive disparity, feature, and degree. */
7  $\mathcal{G}_n(v) \leftarrow$  top- $k$  nodes by  $Score(u)$ 
8  $\hat{\phi} \leftarrow (M^\top U M)^{-1} M^\top U \hat{y}$ 
9 foreach edge  $e = (u, w)$  where  $u, w \in \mathcal{G}_n(v)$  do
10   Assign attribution  $\phi_e$  from  $\hat{\phi}$ 
11  $\mathcal{E}' \leftarrow \mathcal{E} \setminus$  top- $k$  edges by  $\phi_e$ 
12  $\mathcal{G}' \leftarrow (\mathcal{V}, \mathcal{E}', \mathcal{X}, \mathcal{T}')$ 
13 return  $\mathcal{G}'$ 

```
